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# Safety data sheet



BASF Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)
Date / Revised: 10.06.2025 Version: 2.0
Product: **US-600M**
(ID no. 30740149/SDS_GEN_00/EN)
Date of print 28.06.2025

## 1. Identification

### Product identifier

## US-600M

Uses advised against: Consumer spray applications, Consumer applications requiring temperartures above 40°C, Professional cleaning applications with aprotic polar solvents (meeting the IUPAC definition)
Recommended use: polyurethane component, industrial chemicals

### Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Division Monomers

Telephone: +49 621 60 42737
E-mail address: pss.monomers@basf.com

### Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

## 2. Hazards Identification

### Classification of the substance or mixture

According to UN GHS criteria

Acute Tox. 4 (Inhalation - mist)
Skin Irrit. 2
Eye Irrit. 2A
Resp. Sens. 1
Skin Sens. 1B
Carc. 2
STOT SE 3 (irritating to respiratory system)
STOT RE (Olfactory organs) 2 (by inhalation)

For the classifications not written out in full in this section the full text can be found in section 16.

## Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Danger

Hazard Statement:

| | |
|---|---|
| H319 | Causes serious eye irritation. |
| H315 | Causes skin irritation. |
| H332 | Harmful if inhaled. |
| H334 | May cause allergy or asthma symptoms or breathing difficulties if inhaled. |
| H317 | May cause an allergic skin reaction. |
| H335 | May cause respiratory irritation. |
| H351 | Suspected of causing cancer. |
| H373 | May cause damage to organs (Olfactory organs) through prolonged or repeated exposure (inhalation). |

Precautionary Statements (Prevention):

| | |
|---|---|
| P280 | Wear protective gloves, protective clothing and eye protection or face protection. |
| P271 | Use only outdoors or in a well-ventilated area. |
| P260 | Do not breathe dust/gas/mist/vapours. |
| P201 | Obtain special instructions before use. |
| P202 | Do not handle until all safety precautions have been read and understood. |
| P284 | In case of inadequate ventilation wear respiratory protection. |
| P272 | Contaminated work clothing should not be allowed out of the workplace. |
| P264 | Wash contaminated body parts thoroughly after handling. |

Precautionary Statements (Response):

| | |
|---|---|
| P312 | Call a POISON CENTER or physician if you feel unwell. |
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P304 + P340 | IF INHALED: Remove person to fresh air and keep comfortable for breathing. |
| P308 + P313 | IF exposed or concerned: Get medical attention. |
| P302 + P352 | IF ON SKIN: Wash with plenty of soap and water. |
| P333 + P313 | If skin irritation or rash occurs: Get medical attention. |
| P342 + P311 | If experiencing respiratory symptoms: Call a POISON CENTER or doctor/physician. |
| P337 + P313 | If eye irritation persists: Get medical attention. |
| P362 + P364 | Take off contaminated clothing and wash it before reuse. |

Precautionary Statements (Storage):

| | |
|---|---|
| P403 + P233 | Store in a well-ventilated place. Keep container tightly closed. |
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Labeling of special preparations (GHS):
Contains isocyanates. May produce an allergic reaction.

### Other hazards

According to UN GHS criteria

The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative).

Avoid any contact with the substance in case of known allergy to isocyanates, skin complaints, hypersensitivity reactions, chronic respiratory disease, asthmatic attacks or bronchial attacks

## 3. Composition/Information on Ingredients

### Substances

Not applicable

### Mixtures

Chemical nature

Preparation based on: 4,4'-methylenediphenyl diisocyanate; diphenylmethane-4,4'-di-isocyanate

Hazardous ingredients (GHS)
According to UN GHS criteria

4,4'-Methylenediphenyl diisocyanate

| | |
|---|---|
| Content (W/W): >= 50 % - < 75 %<br>CAS Number: 101-68-8<br>EC-Number: 202-966-0<br>INDEX-Number: 615-005-00-9 | Acute Tox. 4 (Inhalation - mist)<br>Skin Irrit. 2<br>Eye Irrit. 2A<br>Resp. Sens. 1<br>Skin Sens. 1B<br>Carc. 2<br>STOT SE 3 (irr. to respiratory syst.)<br>STOT RE (Respiratory system) 2<br>H319, H315, H332, H334, H317, H335, H351, H373<br><br><u>Specific concentration limit:</u><br>Eye Irrit. 2: >= 5 %<br>Skin Irrit. 2: >= 5 %<br>Resp. Sens. 1: >= 0,1 %<br>STOT SE 3, irr. to respiratory syst.: >= 5 % |

Isocyanic acid, polymethylenepolyphenylene ester (P-MDI)

| | |
|---|---|
| Content (W/W): >= 20 % - < 25 %<br>CAS Number: 9016-87-9 | Acute Tox. 4 (Inhalation - mist)<br>Skin Corr./Irrit. 2<br>Eye Dam./Irrit. 2B<br>Resp. Sens. 1<br>Skin Sens. 1B<br>Carc. 2<br>STOT SE 3 (irr. to respiratory syst.)<br>STOT RE (Olfactory organs) 2 (by inhalation)<br>H320, H315, H332, H334, H317, H335, H351, H373 |

Methylenediphenyl diisocyanate

| | |
|---|---|
| Content (W/W): >= 15 % - < 20 %<br>CAS Number: 26447-40-5<br>EC-Number: 247-714-0<br>INDEX-Number: 615-005-00-9 | Acute Tox. 4 (Inhalation - mist)<br>Skin Irrit. 2<br>Eye Irrit. 2A<br>Resp. Sens. 1<br>Skin Sens. 1B<br>Carc. 2<br>STOT SE 3 (irr. to respiratory syst.)<br>STOT RE (Respiratory system) 2<br>H319, H315, H332, H334, H317, H335, H351, H373<br><br><u>Specific concentration limit:</u><br>Skin Irrit. 2: >= 5 %<br>Resp. Sens. 1: >= 0,1 %<br>STOT SE 3, irr. to respiratory syst.: >= 5 %<br>Eye Irrit. 2: >= 5 % |

Isocyanic acid, polymethylenepolyphenylene ester, polymer with.alpha.-hydro-.omega.-hydroxypoly(oxy-1,2-ethanediyl)

| | |
|---|---|
| Content (W/W): >= 0,3 % - < 3 %<br>CAS Number: 57636-09-6 | Skin Corr./Irrit. 2<br>Eye Dam./Irrit. 2A<br>Skin Sens. 1<br>H319, H315, H317 |

2,4-Dioxo-1,3-diazetidine-1,3-diylbis[p-phenylenemethylene-p-phenylene] diisocyanate

| | |
|---|---|
| Content (W/W): >= 0,3 % - < 1 %<br>CAS Number: 17589-24-1<br>EC-Number: 241-559-2 | Acute Tox. 4 (Inhalation - vapour)<br>Skin Irrit. 2<br>Eye Irrit. 2A<br>Resp. Sens. 1<br>Skin Sens. 1<br>Carc. 2<br>STOT SE 3 (irr. to respiratory syst.)<br>STOT RE (Respiratory system) 2<br>Aquatic Acute 2<br>H319, H315, H332, H334, H317, H335, H351, H373, H401<br>EUH204 |

For the classifications not written out in full in this section the full text can be found in section 16.

## 4. First-Aid Measures

### Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

### Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., Eye irritation, skin irritation, allergic symptoms

Hazards: Symptoms can appear later.

### Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote, administer corticosteroid dose aerosol to prevent pulmonary odema.

## 5. Fire-Fighting Measures

### Extinguishing media

Suitable extinguishing media:
dry powder, carbon dioxide, alcohol-resistant foam, water spray

### Special hazards arising from the substance or mixture

Carbon dioxide, Carbon monoxide, Hydrogen cyanide, nitrogen oxides
The substances/groups of substances mentioned can be released in case of fire.

### Advice for fire-fighters

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Keep containers cool by spraying with water if exposed to fire. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

## 6. Accidental Release Measures

### Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Ensure adequate ventilation. Use breathing apparatus if exposed to vapours/dust/aerosol.

### Environmental precautions

Do not empty into drains. Do not discharge into the subsoil/soil.

### Methods and material for containment and cleaning up

For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

## 7. Handling and Storage

### Precautions for safe handling

Provide suitable exhaust ventilation at the processing machines. Ensure thorough ventilation of stores and work areas. Avoid aerosol formation. When handling heated product, vapours of the product should be ventilated, and respiratory protection used. Wear respiratory protection when spraying. Danger of bursting when sealed gastight. Protect against moisture. Clean up contamination as soon as they occur. Provide basic employee training to prevent/minimize exposures. Products freshly manufactured from isocyanates can contain incompletely reacted isocyanates and other dangerous substances, e.g. primary aromatic amines. Industrial cleaning with aprotic polar solvents

(meeting the IUPAC definition) may lead to formation of hazardous primary aromatic amine (>0,1%). See Section 11.

Protection against fire and explosion:
No special precautions necessary.

### Conditions for safe storage, including any incompatibilities

Keep away from water. Segregate from foods and animal feeds. Segregate from acids and bases.

Suitable materials for containers: Carbon steel (Iron), High density polyethylene (HDPE), Low density polyethylene (LDPE), Stainless steel 1.4301 (V2)
Further information on storage conditions: From this information no suitability of the materials mentioned above for the design of installations, including containers for permanent storage, can be inferred. Special conditions apply to the selection of materials in this regard, which we can communicate on request.
Keep container tightly closed in a cool, well-ventilated place. Protect against moisture. Formation of CO2 and build up of pressure possible. Danger of bursting when sealed gastight.

Storage stability:
Storage temperature: 27 - 35 °C
Protect against moisture.
The stated storage temperature is noted for health and safety in the workplace. With regard to Quality, please refer to the product specific Technical Bulletin.

### Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

## 8. Exposure Controls/Personal Protection

### Control parameters

Components with occupational exposure limits

101-68-8: 4,4'-Methylenediphenyl diisocyanate
17589-24-1: 2,4-Dioxo-1,3-diazetidine-1,3-diylbis[p-phenylenemethylene-p-phenylene] diisocyanate
26447-40-5: Methylenediphenyl diisocyanate
9016-87-9: Isocyanic acid, polymethylenepolyphenylene ester (P-MDI)

### Exposure controls

Personal protective equipment

Respiratory protection:
Respiratory protection in case of vapour/aerosol release. Combination filter for gases/vapours of organic, inorganic, acid inorganic and alkaline compounds (e.g. EN 14387 Type ABEK).

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
butyl rubber (butyl) - 0.7 mm coating thickness
nitrile rubber (NBR) - 0.4 mm coating thickness
chloroprene rubber (CR) - 0.5 mm coating thickness
Unsuitable materials
polyvinylchloride (PVC) - 0.7 mm coating thickness
Polyethylene-Laminate (PE laminate) - ca. 0.1 mm coating thickness

Suitable materials that provide sufficient protection for industrial cleaning with aprotic polar solvents (meeting the IUPAC definition):
butyl rubber (butyl) - 0.7 mm coating thickness
nitrile rubber (NBR) - 0.4 mm coating thickness
chloroprene rubber (CR) - 0.5 mm coating thickness

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Do not breathe vapour/spray. With products freshly manufactured from isocyanates body protection and chemical resistant protective gloves is recommended. Wearing of closed work clothing is required additionally to the stated personal protection equipment. No eating, drinking, smoking or tobacco use at the place of work. Take off immediately all contaminated clothing. Hands and/or face should be washed before breaks and at the end of the shift. At the end of the shift the skin should be cleaned and skin-care agents applied.

## 9. Physical and Chemical Properties

### 9.1. Information on basic physical and chemical properties

| | | |
|---|---|---|
| State of matter: | liquid | |
| Form: | liquid | |
| Colour: | dark brown | |
| Odour: | faintly aromatic | |
| Odour threshold: | not applicable | |
| Freezing point: | 11,00 °C | |
| Melting point: | No data available. | |
| Boiling point: | 200,00 °C<br>(5,000000 mmHg) | |
| Sublimation point: | No applicable information available. | |
| Flammability: | not flammable | (derived from flash point) |
| Lower explosion limit: | For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point. | |

Upper explosion limit:
For liquids not relevant for classification and labelling.

Flash point: > 200,00 °C (open cup)

Auto-ignition temperature: > 470,00 °C

Thermal decomposition: No decomposition if stored and handled as prescribed/indicated.

pH value:
not applicable

Viscosity, kinematic:
No applicable information available.

Viscosity, dynamic: 33,000 mPa.s
(25,00 °C)

Solubility in water: Reacts with water.

Solubility (qualitative) :
No applicable information available.

Solubility (quantitative):
No applicable information available.

Partitioning coefficient n-octanol/water (log Kow):
Unspecified

Vapour pressure: 0,00001 mmHg
(25,00 °C)

Relative density:
No applicable information available.

Density: 1,2200 g/cm3
(25,00 °C)

Relative vapour density (air):
not applicable

### 9.2. Other information

**Information with regard to physical hazard classes**

Oxidizing properties
Fire promoting properties: Not an oxidizer.
Self-heating substances and mixtures
Self heating ability: not applicable, the product is a liquid
Corrosion to metals

No corrosive effect on metal.

**Other safety characteristics**

Miscibility with water:
Reacts with water.

Other Information: If necessary, information on other physical and chemical parameters is indicated in this section.

Evaporation rate:
Value can be approximated from Henry's Law Constant or vapor pressure.

## 10. Stability and Reactivity

### Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: No corrosive effect on metal.

### Possibility of hazardous reactions

On contact with water, gaseous decomposition products are formed, which cause build-up of pressure in tightly closed containers. Risk of bursting. Reacts with substances which contain active hydrogen.

### Conditions to avoid

Avoid moisture.

### Incompatible materials

Substances to avoid:
acids, alcohols, amines, water, Alkalines

### Hazardous decomposition products

:
No hazardous decomposition products if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Of moderate toxicity after short-term inhalation. Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact.

Experimental/calculated data:
LD50 rat (oral): > 10.000 mg/kg

LC50 rat (by inhalation): > 2,24 mg/l 1 h (OECD Guideline 403)
An aerosol was tested.

LD50 rabbit (dermal): > 10.000 mg/kg

Irritation

Assessment of irritating effects:
Irritating to eyes and skin.

Experimental/calculated data:
Skin corrosion/irritation: Irritant.

Serious eye damage/irritation: Irritant.

Respiratory/Skin sensitization

Assessment of sensitization:
The substance may cause sensitization of the respiratory tract. Sensitization after skin contact possible. Studies in animals suggest that dermal exposure may lead to pulmonary sensitization. However, the relevance of this result for humans is unclear.

Germ cell mutagenicity

Assessment of mutagenicity:
Results from a number of mutagenicity studies with microorganisms, mammalian cell culture and mammals are available. Taking into account all of the information, there is no indication that the substance is mutagenic.

Carcinogenicity

Assessment of carcinogenicity:
A carcinogenic effect cannot safely be ruled out.

Industrial cleaning with aprotic polar solvents (meeting the IUPAC definition) may lead to formation of hazardous primary aromatic amine (>0,1%). Primary aromatic amines are chemicals that are regarded as potenttially carcinogenic for humans based on animal testing. Some of these chemicals are known human carcinogens. No adverse health effects are anticipated if recommended personal protective equipment and industrial hygiene practices are used.

Reproductive toxicity

Assessment of reproduction toxicity:
No effects have been reported in reproductive organs in long term animal studies. Study scientifically not justified.

Developmental toxicity

Assessment of teratogenicity:
The substance did not cause malformations in animal studies; however, toxicity to development was observed at high doses that were toxic to the parental animals.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Causes temporary irritation of the respiratory tract.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
After repeated exposure the prominent effect is local irritation. The substance may cause damage to the olfactory epithelium after repeated inhalation. These effects are not relevant to humans at occupational levels of exposure.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

The product has not been tested. The statement has been derived from the properties of the individual components.

## 12. Ecological Information

### Toxicity

Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.
The product may hydrolyse. The test result maybe partially due to degradation products. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Toxicity to fish:
LC0 (96 h) > 1.000 mg/l, Brachydanio rerio (OECD Guideline 203, static)

Aquatic invertebrates:
EC50 (24 h) > 1.000 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)

Aquatic plants:
EC0 (72 h) 1.640 mg/l (growth rate), Scenedesmus subspicatus (OECD Guideline 201, static)

Microorganisms/Effect on activated sludge:
EC50 (3 h) > 100 mg/l, activated sludge, domestic, aerobic (OECD Guideline 209, aquatic)

### Persistence and degradability

Assessment biodegradation and elimination (H2O):
Poorly biodegradable. The product is unstable in water. The elimination data also refer to products of hydrolysis.

Elimination information:
0 % BOD of the ThOD (28 d) (OECD Guideline 302 C) (aerobic, activated sludge) Poorly biodegradable.

Assessment of stability in water:
In contact with water the substance will hydrolyse slowly.
Information on Stability in Water (Hydrolysis):
$t_{1/2}$ 20 h (25 °C)

### Bioaccumulative potential

Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

Bioaccumulation potential:
Bioconcentration factor: 200 (28 d), Cyprinus carpio (OECD Guideline 305 E)

### Mobility in soil

Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is not expected.

### Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

### Additional information

Adsorbable organically-bound halogen (AOX):
This product contains no organically-bound halogen.

## 13. Disposal Considerations

### Waste treatment methods

Incinerate in suitable incineration plant, observing local authority regulations.
Dispose of isocyanate waste in dry containers and never mix together with other wastes (reaction, dangerous pressure build up).

Waste key:
08 05 01¤ waste isocyanates

Contaminated packaging:
Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

## 14. Transport Information

Transport in inland waterway vessel
Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |

| | |
|---|---|
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Air transport**

IATA/ICAO

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

### Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

## 15. Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

## 16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

| | |
|---|---|
| Acute Tox. | Acute toxicity |
| Skin Irrit. | Skin irritation |
| Eye Irrit. | Eye irritation |
| Resp. Sens. | Respiratory sensitization |
| Skin Sens. | Skin sensitization |
| Carc. | Carcinogenicity |
| STOT SE | Specific target organ toxicity — single exposure |
| STOT RE | Specific target organ toxicity — repeated exposure |
| Skin Corr./Irrit. | Skin corrosion/irritation |
| Eye Dam./Irrit. | Serious eye damage/eye irritation |
| Aquatic Acute | Hazardous to the aquatic environment - acute |
| H319 | Causes serious eye irritation. |
| H315 | Causes skin irritation. |
| H332 | Harmful if inhaled. |
| H334 | May cause allergy or asthma symptoms or breathing difficulties if inhaled. |
| H317 | May cause an allergic skin reaction. |
| H335 | May cause respiratory irritation. |

| | |
|---|---|
| H351 | Suspected of causing cancer. |
| H373 | May cause damage to organs (Respiratory system) through prolonged or repeated exposure. |
| H320 | Causes eye irritation. |
| H373 | May cause damage to organs (Olfactory organs) through prolonged or repeated exposure (inhalation). |
| H401 | Toxic to aquatic life. |
| EUH204 | Contains isocyanates. May produce an allergic reaction. |

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.